# Fiscal Policy in General Equilibrium: Model Specification and Worked Cases

A companion to the Baxter & King (1993) replication app

**Abstract**

This document is written for undergraduates using the interactive app at `https://profguclass.github.io/baxter-king-fiscal-app/` to explore Baxter & King (1993, *American Economic Review*), "Fiscal Policy in General Equilibrium." Part I lays out the full model the app solves: preferences, technology (including the Section VI extension for productive public capital), the government's budget, the household's optimality conditions, the closed-form steady state, the log-linear transition dynamics, and how the app's four sections (S1–S4, in two parts) map onto that math. Part II works through seven cases that are easy to get wrong intuitively — why a spending *composition* change is not economically neutral even when public capital is unproductive; why the *financing rule* moves the economy even when every dollar of tax revenue is handed straight back; why several steady-state variables always move by the exact same percentage by mathematical necessity, not by coincidence; why the app can lose its saddle path altogether at extreme income-tax rates; why a "temporary" spending increase does not bring the economy back to where it started once the underlying policy regime has changed; why the real interest rate shapes fiscal multipliers even though it looks like a completely separate parameter; and why longer-lasting temporary shocks produce *bigger*, not smaller, effects on impact. Part III derives every first-order condition and steady-state equation from scratch.

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# Part I
# Model specification

## 1 Notation

| Symbol | Meaning |
|---|---|
| $C_t$ | Consumption |
| $N_t$ | Labor supplied (fraction of the time endowment); $L_t = 1 - N_t$ is leisure |
| $Y_t$ | Output |
| $K_t$ | Private capital stock |
| $K_t^G$ | Public (government) capital stock |
| $I_t,\ I_t^G$ | Private and public investment |
| $G_t^T$ | Lump-sum government transfers |
| $w_t,\ r_t$ | Real wage and real rental rate on capital |
| $\tau_t$ | Tax rate |
| $\theta_N,\ \theta_K = 1 - \theta_N$ | Labor's and capital's shares of income |
| $\theta_G$ | Productivity of public capital (Section VI extension) |
| $\theta_L$ | Weight on leisure in utility (calibrated, not a free parameter) |
| $\delta$ | Depreciation rate (same for $K$ and $K^G$) |
| $A$ | Total factor productivity, normalized to 1 |

A hat over a variable, e.g. $\hat{y}_t$, denotes a percent deviation from steady state — the units used in every transition-path chart in the app.

## 2 Preferences

$$u(C_t, N_t) = \ln C_t + \theta_L \ln(1 - N_t) \tag{1}$$

Households value consumption and leisure, with log utility in each — the standard King–Plosser–Rebelo specification that keeps balanced-growth labor supply well behaved. Notice that government spending does *not* appear here at all: households get no direct utility from transfers $G_t^T$ or public investment $I_t^G$. This single fact is the seed of almost everything counterintuitive in Part II — government spending only ever affects the household *indirectly*, through the resource constraint and the budget constraint, never through a warm-glow "I like government spending" term.

**Why is there no "basic government purchases" category?** Many textbook treatments split $G$ into wasteful "purchases" $G_B$, productive investment $I^G$, and transfers $G^T$. Here $G_B$ is dropped because it would be economically identical to $I^G$ at $\theta_G = 0$: neither enters utility, and both use up real resources one-for-one. Keeping a separate $G_B$ category would just be relabeling. Total government spending in this model is therefore always exactly two pieces: $G = I^G + G^T$.

## 3 Technology

$$Y_t = A\, K_t^{\theta_K}\, (K_t^G)^{\theta_G}\, N_t^{\theta_N}, \qquad \theta_K = 1 - \theta_N \tag{2}$$

Output is Cobb–Douglas in private capital and labor, exactly as in the paper's baseline model. The Section VI extension (and the app's $\theta_G$ slider) adds public capital $K_t^G$ as a third input: it raises the productivity of every private input, without the government paying for it again next period — a non-rival "public good" flavor. Setting $\theta_G = 0$ collapses this exactly to the paper's baseline model, where public investment is a pure resource cost with *no* productivity effect. Note that $\theta_K + \theta_N = 1$ (constant returns to scale in $(K, N)$ alone) but $\theta_G$ is *extra*: when $\theta_G > 0$ the production function has *increasing* returns to scale in $(K, K^G, N)$ jointly, which is exactly the source of the model's "multiplier $> 1$" results in Part II.

## 4 Capital accumulation

$$K_{t+1} = (1-\delta)K_t + I_t \qquad\qquad K_{t+1}^G = (1-\delta)K_t^G + I_t^G \tag{3}$$

Both stocks accumulate by the standard perpetual-inventory rule and (for simplicity) depreciate at the same rate $\delta$.

## 5 Resource constraint and the government budget

$$Y_t = C_t + I_t + I_t^G \tag{4}$$

Output is used for private consumption, private investment, and public investment. **Transfers do not appear here.** A transfer just moves resources from the government's books to a household's pocket without using up any output, so it nets out of the economy-wide resource constraint even though it is part of the government's budget below.

$$\tau_t Y_t = I_t^G + G_t^T, \qquad \tau_t = \frac{I_t^G + G_t^T}{Y_t} \tag{5}$$

The government's budget balances every period (no debt in this model), and the tax rate is *defined* as total government spending divided by output. This holds identically under *both* financing rules below — what differs between them is not how much revenue is raised, but whether raising it distorts the household's decisions.

$$\text{Income tax:} \quad C_t + I_t = (1-\tau_t)\big(w_t N_t + r_t K_t\big) + G_t^T \tag{6}$$

$$\text{Lump-sum:} \quad C_t + I_t = w_t N_t + r_t K_t - \tau_t Y_t + G_t^T \tag{7}$$

Under **income tax** financing, $(1-\tau_t)$ is a proportional wedge that shrinks the *marginal* return to working and saving — this is what makes the tax distortionary. Under **lump-sum** financing, the same total revenue $\tau_t Y_t$ is instead collected as a poll tax that does not depend on how much the household works or saves, so the household faces $w_t$ and $r_t$ directly (no $(1-\tau_t)$ term at all): a pure income (wealth) effect with no substitution effect at the margin.

## 6 Household optimization

Maximizing lifetime utility subject to the budget constraint gives two first-order conditions, one per financing rule.

$$\text{Labor-leisure, income tax:} \qquad \theta_L \frac{C_t}{1-N_t} = (1-\tau_t)\, w_t \tag{8}$$

$$\text{Labor-leisure, lump-sum:} \qquad \theta_L \frac{C_t}{1 - N_t} = w_t \tag{9}$$

The household works until the marginal disutility of an extra hour (left side) equals its marginal after-tax benefit (right side). Under income tax financing that benefit is shrunk by $(1 - \tau_t)$; under lump-sum financing there is no such wedge at all.

$$\text{Euler, income tax:} \qquad \frac{1}{C_t} = \beta \frac{1 + (1 - \tau_{t+1})\,\text{MPK}_{t+1} - \delta}{C_{t+1}} \tag{10}$$

$$\text{Euler, lump-sum:} \qquad \frac{1}{C_t} = \beta \frac{1 + \text{MPK}_{t+1} - \delta}{C_{t+1}}, \qquad \beta = \frac{1}{1 + r} \tag{11}$$

There is no expectation operator here: this model has no stochastic element (no random shocks to technology, policy, or anything else), so every "future" variable is simply its known, perfect-foresight value. The household is not *guessing* what $C_{t+1}$ or $\text{MPK}_{t+1}$ will be — under a given policy path it *knows*, and the Euler equation is an ordinary deterministic difference equation linking today's consumption to tomorrow's. (This is what makes the transition path in Section 7 below solvable by exact saddle-path methods rather than by simulation.)

Prices are marginal products: $w_t = \theta_N Y_t / N_t$ and $\text{MPK}_t = \theta_K Y_t / K_t$. In steady state, $(1 - \tau)\text{MPK} = r + \delta$ under income tax, and $\text{MPK} = r + \delta$ under lump-sum.

# 7 Steady state (closed form)

Because $\theta_K + \theta_N = 1$, dividing the technology equation through by $N$ leaves the capital/labor ratio $\kappa = K/N$ pinned down by a capital-Euler condition written in *per-worker* public capital $K^G/N$ — but with an extra $N^{\theta_G}$ factor that does *not* cancel, since $K^G$ is itself funded as a share of *aggregate* output, not of output per worker:

$$\text{Income tax:} \qquad (1 - \tau)\,\theta_K\, A\,(K^G/N)^{\theta_G}\, N^{\theta_G}\, \kappa^{\theta_K - 1} = r + \delta \tag{12}$$

$$\text{Lump-sum:} \qquad \theta_K\, A\,(K^G/N)^{\theta_G}\, N^{\theta_G}\, \kappa^{\theta_K - 1} = r + \delta \tag{13}$$

When $\theta_G = 0$ the $N^{\theta_G}$ factor is just 1 and this collapses to the familiar CRS-in-$(K, N)$ formula, solved for $\kappa$ directly. When $\theta_G > 0$, $K^G = I^G/\delta$ itself depends on output, *and* the equation above depends on $N$ directly too, so the app solves a small nested fixed point (first on $\kappa$ given a guess for $N$, then updating $N$, iterating to convergence).

Everything else then follows in closed form, with $s_C = 1 - s_I - s_{IG}$ the steady-state consumption share (transfers never reduce $s_C$, since they are not resource-using) and $s_I = \delta\kappa/(Y/N)$ the investment share:

$$\text{Income tax:} \qquad N = \frac{(1 - \tau)\,\theta_N}{\theta_L\, s_C + (1 - \tau)\,\theta_N} \tag{14}$$

$$\text{Lump-sum:} \qquad N = \frac{\theta_N}{\theta_L\, s_C + \theta_N} \tag{15}$$

$$Y = \frac{Y}{N} \cdot N, \qquad K = \kappa N, \qquad C = s_C\, Y, \qquad I = s_I\, Y \tag{16}$$

$\theta_L$ itself is not a free parameter: it is *calibrated* once, at a chosen baseline policy, so that steady-state $N$ matches a target hours-worked share (the paper's Table 1 strategy, and $N_{\text{target}} = 20\%$ in the app). Holding that calibrated $\theta_L$ fixed while comparing two different policies is what lets $N$ move *honestly* in response to a policy change, rather than being pinned back to the target by construction.

# 8 Transition dynamics (log-linearized)

Around a steady state, every equation above can be approximated to first order in percent deviations. The production and labor-supply conditions combine into one static relationship,

$$\hat{y}_t = y_k\,\hat{k}_t + y_{kg}\,\hat{k}_t^G + y_c\,\hat{c}_t + y_g\,\hat{g}_t, \tag{17}$$

capital accumulation becomes a linear law of motion for the *predetermined* state $\hat{k}_t$,

$$\hat{k}_{t+1} = \Phi_{kk}\,\hat{k}_t + \Phi_{kc}\,\hat{c}_t + \Phi_{kg}\,\hat{g}_t + \Phi_{k,kg}\,\hat{k}_t^G, \tag{18}$$

and the Euler equation pins down the *forward-looking* ("jump") variable $\hat{c}_t$ in terms of tomorrow's capital and government spending,

$$\hat{c}_t = \text{coef}_c\,\hat{c}_{t+1} + \text{coef}_k\,\hat{k}_{t+1} + \text{coef}_g\,\hat{g}_{t+1} + \text{coef}_{kg}\,\hat{k}_{t+1}^G. \tag{19}$$

Public capital's own law of motion is exogenous, so the system has exactly one predetermined and one jump variable: it is **saddle-path stable**, with one stable and one unstable eigenvalue of the reduced-form transition matrix. The app diagonalizes that matrix and pins the unstable component to the present value of *future* government-spending shocks (the King–Plosser–Rebelo / Blanchard–Kahn method) rather than simulating forward naively, which would blow up numerically.

# 9 Reading the app: two parts, four sections

Every chart in the app is one of the two objects above: the closed-form steady state of Section 7, or the log-linear transition path of this section. The app presents them in **two parts with four sections**. The two parts differ only in *which two steady states are being compared* — the underlying mathematics is the same in both. Part I compares the sidebar's current settings with a fixed benchmark (S1 and S2); Part II takes the current settings as given and raises $G/Y$ from there (S3 and S4).

The sidebar items are lettered: A, the tax mix (each tax's share of revenue); B, the productivity of public capital $\theta_G$; C, the value of in-kind welfare to households $\varphi$; D, total government spending (D.1 the current $G/Y$, D.2 the further increase used in Part II, D.3 whether a change in $G/Y$ is permanent or temporary); E, the composition of $G$ (public investment, in-kind welfare, cash welfare); and F, the share $\lambda$ of hand-to-mouth households. The real interest rate is fixed at $r = 6.5\%$, with $\theta_N = 0.58$ and $\delta = 10\%$ as in Baxter & King.

## 9.1 Part I, S1: Benchmark vs Now

Compares a *benchmark* economy against **whatever the sidebar is currently set to**. By default the benchmark is the app's starting point: tax revenue 50% lump-sum, 20% income tax and 30% consumption tax; $\theta_G = 0.10$; $G/Y = 20\%$, split 20% public investment, 50% in-kind welfare and 30% cash welfare; $\varphi = 0.10$; and $\lambda = 0.30$. The "Benchmark" button lets a user replace it for the rest of the session. Every control can differ between the two columns at once, so this section reports the *net, cumulative* effect of everything that has been changed. It is the right place for questions like Cases A and B below: "what does a different tax mix do to the economy, holding everything else at the benchmark?"

Its steady-state comparison is the direct application of Section 7: solve for the benchmark's $(\kappa, N, Y, C, \ldots)$ once, solve for the current settings' $(\kappa, N, Y, C, \ldots)$ once, and report percent differences. Both sides re-use one calibrated $\theta_L$ (and one reference level of public capital), fixed at

the default benchmark even when a user sets their own. Labor supply is therefore free to respond honestly rather than being pinned back to the same target by construction, and the same settings always describe the same economy.

Its *time-series* chart applies the log-linear transition machinery of this section in a way that goes beyond a simple $\Delta G/Y$ experiment. The tax mix, composition, $\theta_G$, $\varphi$ and $\lambda$ are always treated as a **permanent, immediate switch in regime** — this model has no mechanism for a structural parameter to "revert," so there is no sensible "temporary $\theta_G$" or "temporary tax mix." The economy is imagined to start ($t = 0$) with the *benchmark's* capital stocks $K, K^G$ but the *current settings'* parameters and policy from that point on, with the transition linearized around the *destination* steady state:

$$\hat{k}_0 = \ln\frac{K_{\text{benchmark}}}{K_{\text{current}}}, \qquad \hat{k}_0^G = \ln\frac{K^G_{\text{benchmark}}}{K^G_{\text{current}}}. \tag{20}$$

Only total government spending $G$ respects the Duration control (item D.3), since a temporary spending change is economically meaningful even when the surrounding policy regime is not. If Permanent, $\hat{g}_t = 0$ for all $t$ ($G/Y$ jumps to the current share immediately and stays there). If Temporary, $G/Y$ instead reverts to the *benchmark's* share after the chosen number of years — see Case E below for why this is more subtle than it first looks.

### 9.2 Part I, S2: Growth, inequality, and welfare over time

Follows the *same* transition path as S1, year by year, and asks four questions: does the economy grow, whose income grows, does inequality between the two household types rise or fall, and are households better or worse off? Its tabs (Growth, Incomes, Hours, Inequality, Welfare) report growth of output and labor productivity; labor, capital and disposable income and hours by household type; income and consumption ratios and a two-group Gini; and consumption-equivalent welfare gains, year by year and overall, with and without the transition. Rather than reading these off the linear approximation, S2 solves each year's static equations exactly at S1's path of capital and optimizers' consumption, so for large policy changes the first years can differ a little from S1's charts; both end at the same steady state. Every chart has a data table under it with a final row for the cumulative effect over the years shown.

Everything in these tabs follows the "Years to display" slider (50 years by default, i.e. years 0–49). The Inequality tab opens with a summary of the income ratio, the consumption ratio and the Gini in year $-1$ and in the last year shown, and the change between them. The Welfare tab adds up discounted utility over the years shown and holds the last year shown forever; "long run only" uses that last year alone. Both are reported as consumption-equivalent gains (%), and the labels name the years used.

### 9.3 Part II, S3: Now vs After raising G/Y, in levels and over time

Starts from **the sidebar's current settings**, including the current $G/Y$ (item D.1), and raises $G/Y$ by the amount in item D.2, holding the tax mix, composition, $\varphi$, $\theta_G$ and $\lambda$ fixed at their current values *on both sides* of the comparison. This isolates the effect of more government spending in whatever economy the sidebar describes: change any other setting and the starting point changes, so these effects change too. The left column is the same steady state as S1's "Current settings."

Its time series is the ordinary $\Delta G/Y$ transition path: $\hat{g}_t$ is the (possibly temporary) government-spending shock itself, $\hat{k}_0 = 0$ (capital starts exactly at the pre-shock steady state, since only $G/Y$ changed, nothing structural), and the linearization is around whichever steady state the economy

ends up in (the new steady state, for a permanent shock; the current one, for a temporary shock, after which $G/Y$ returns to its current level).

The table's right column is always the steady state of a *permanent* increase, so item D.3 (Duration) changes the transition path but not the table. With a temporary increase the economy never reaches that column: once $G/Y$ drops back, the path returns to the left column. The app says so in a note under the table whenever D.3 is Temporary.

The transition path has five tabs. Commodity, Labor and Financial market show the % deviations above. Inequality and Welfare work exactly like S2's tabs, with the current situation in place of the benchmark: each year's static equations are solved exactly along this path, year $-1$ is the current situation, and welfare gains are measured against staying there.

### 9.4 Part II, S4: Now vs After raising G/Y, per dollar of new spending

Takes the *same* pair of steady states as S3 — the current $G/Y$ vs. the current $G/Y$ plus the item D.2 increase, everything else held fixed — and reports, for every variable in S3's table (output, consumption, investment, both capital stocks, $G$ and its three parts, hours, the wage, both tax rates and the lump-sum tax; and, when $\lambda > 0$, each household type's total consumption and hours), the multiplier

$$\frac{\Delta X}{\Delta G} = \frac{X_{\text{after}} - X_{\text{now}}}{G_{\text{after}} - G_{\text{now}}}, \tag{21}$$

both **long-run** (using the two steady-state levels directly) and **on impact** (using each variable's year-0 level along S3's own transition path in place of $X_{\text{after}}$). No new steady state or transition path is computed here — it is purely S3's comparison, in dollars per dollar of new spending rather than percent changes. Hours and tax rates are in percentage points per unit of $G$; their numbers are much larger, so they get a second bar chart. The tax rates and the lump-sum tax have no linearized path, so their year-0 levels come from solving year 0's static equations exactly. As in S3, the long-run column always describes a permanent increase: under a temporary one the economy returns to the current situation, so only the impact column depends on item D.3 (the app notes this when D.3 is Temporary). A multiplier above $+1$ for output means a marginal dollar of spending raises long-run output by *more* than a dollar, the paper's central result; investment's impact multiplier can be large even when its long-run multiplier is small, since capital only needs to move once to reach the new steady state (the "accelerator boom").

**Why two parts?** A single "current settings vs. some baseline" comparison cannot serve every purpose at once. If the baseline tracks the current settings on everything except $G/Y$ (Part II's convention), then changing the tax mix *alone* shows up as *no change at all*, since both sides of the comparison use the same tax mix. If instead the baseline is one fixed benchmark (Part I's convention), every control that differs from the benchmark shows up at once, so the effect of more spending alone cannot be read off. Splitting the two removes this ambiguity: S1 and S2 answer "how far is the economy, in total and over time, from the benchmark?"; S3 answers "what, specifically, does more government spending do here?"; and S4 answers the same question as S3, per dollar of new spending.

# Part II
# Worked cases

The cases below are the ones students most often get wrong by intuition alone. Cases A–C follow directly from the closed-form steady state of Section 7; Cases D–E require the transition dynamics of Section 9 as well, and are best explored live in the app rather than by hand.

## 10 Case A: composition is *not* neutral once $I^G > 0$, even at $\theta_G = 0$

**Claim to test:** "If public capital isn't productive ($\theta_G = 0$), then it shouldn't matter whether government spending is transfers or public investment — both are just money moving around."

**Why this is wrong.** Transfers $G^T$ are *not* resource-using: they cancel out of the resource constraint $Y = C + I + I^G$ entirely. Public investment $I^G$ *is* resource-using, with or without $\theta_G > 0$ — it is a real claim on output that competes with $C$ and $I$ for the same $Y$. So shifting the composition of a fixed total $G/Y$ from transfers toward public investment is economically identical to raising ordinary, unproductive government purchases: a negative wealth effect that makes the household work more, even though public capital does nothing for productivity.

**Numerical example.** $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $G/Y = 20\%$, $\theta_G = 0$, lump-sum financing, $\theta_L$ calibrated once at $I^G/G = 0\%$ so that $N = 20.0\%$ there. **S1 (Benchmark vs Now)**, Section 9, shows the same mechanism if only the composition (item E) is moved, starting from the app's own benchmark rather than this example's.

| $I^G/G$ share | $N$ | $Y$ | $C$ | $I^G$ |
|---|---|---|---|---|
| 0% (all transfers) | 20.0% | 0.3934 | 0.2933 | 0.0000 |
| 50% | 22.4% | 0.4407 | 0.2845 | 0.0441 |
| 100% (all investment) | 25.5% | 0.5009 | 0.2732 | 0.1002 |

$N$, $Y$, and $I$ all rise as the investment share rises, purely from the resource-cost channel — *before* any productivity effect is even switched on. This is exactly why, at $\theta_G = 0$, **S4** still reports a nonzero long-run output multiplier $\Delta Y/\Delta G$ for an ordinary, unproductive spending increase ($\approx 1.1$–$1.2$ under lump-sum financing) — $\theta_G = 0$ turns off the *productivity* channel, not the resource-cost channel that drives this table.

**The one truly neutral corner case.** If (and only if) financing is lump-sum *and* the composition is already 100% transfers ($I^G/G = 0$), then changing the *level* of $G/Y$ is a pure wash: a lump-sum tax funding an equal-sized lump-sum transfer leaves $Y, C, I, K, N$ completely flat, since there is no $(1 - \tau)$ wedge to distort anything and no resource-using spending to crowd anything out. This is the *only* combination of settings under which government spending has literally zero real effect.

## 11 Case B: the financing rule moves the economy even at 100% transfers

**Claim to test:** "If every dollar of tax revenue is handed straight back to households as a transfer, it shouldn't matter whether it was collected as a lump-sum poll tax or an income tax — the household

ends up with the same net income either way."

**Why this is wrong.** The claim confuses the *level* of net transfers with the *marginal* incentive to work and save. Under income tax financing, the household's capital-Euler condition contains a $(1-\tau)$ wedge on the *pre-tax* marginal product of capital: $(1-\tau)\,\theta_K A\kappa^{\theta_K-1} = r+\delta$, instead of $\theta_K A\kappa^{\theta_K-1} = r+\delta$. To earn the same after-tax return $r$, the household needs a *higher* pre-tax marginal product, which means holding *less* capital per worker $\kappa$ in equilibrium — a real, permanent effect on the capital stock that has nothing to do with where the tax revenue eventually lands. The same logic applies to the labor-leisure condition.

**Numerical example.** $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $G/Y = 20\%$, all of it transfers ($I^G/G = 0\%$), $\theta_G = 0$. $\theta_L$ is calibrated *separately* under each financing rule so that $N = 20.0\%$ in both cases — this isolates the capital-stock channel from any mechanical change in hours. **S1 (Benchmark vs Now)**, Section 9, shows this same financing switch, but holds $\theta_L$ fixed at the benchmark's calibrated value throughout, so $N$ moves too under income tax (it does not recalibrate back to 20.0%) — the mechanism is identical, only this table's presentation isolates it more cleanly.

| Financing | $N$ | $Y$ | $C$ | $K$ | $w$ |
|---|---|---|---|---|---|
| Lump-sum | 20.0% | 0.3934 | 0.2933 | 1.0014 | 1.1409 |
| Income tax | 20.0% | 0.3347 | 0.2666 | 0.6816 | 0.9707 |

Switching from lump-sum to income tax financing, with $G/Y$, composition, and $\theta_G$ all unchanged, cuts $Y$ by $\approx 15\%$ and $K$ by $\approx 32\%$. $N$ is unchanged only because $\theta_L$ was *recalibrated* to hit the same target hours under each rule — if $\theta_L$ is instead held fixed at one rule's calibrated value (as the app does when comparing two *policies* under the *same* financing rule), $N$ falls too under income tax, since the labor-leisure wedge directly discourages work at the margin.

**Takeaway.** There is no "neutral financing switch" the way there is a neutral composition/level corner case in Case A. The distortion lives on the household's first-order conditions — the *margin* — not on the government's balance sheet, so it cannot be undone by any choice of what the revenue funds.

## 12 Case C: four steady-state variables always move by the identical percentage

**Claim to test:** "If two variables show the exact same percent change in a comparison table, that must be a coincidence or a bug."

**Why this is wrong.** Three of the model's accounting identities are exact, not approximate, and they chain together to force four variables — total government spending $G$, public investment $I^G$, transfers $G^T$, and public capital $K^G$ — to move by *exactly* the same percentage whenever the composition shares and $\delta$ are held fixed between the two points being compared:

$$I^G = f_{I^G}\cdot G, \qquad G^T = f_{G^T}\cdot G \qquad \Longrightarrow \qquad \%\Delta I^G = \%\Delta G^T = \%\Delta G \tag{22}$$

$$K^G = \frac{I^G}{\delta} \qquad \Longrightarrow \qquad \%\Delta K^G = \%\Delta I^G \tag{23}$$

Chaining these: $\%\Delta K^G = \%\Delta I^G = \%\Delta G = \%\Delta G^T$, always, regardless of what is *driving* the change in $G$ (a change in $\theta_G$, $r$, financing, or $G/Y$ itself). The *levels* of these four variables can differ enormously — $K^G$ is typically many times larger than $I^G$, since $K^G = I^G/\delta$ with $\delta$ small — but the *ratio* between any two comparison points is forced to be identical by these three equations, so their percent changes print identically in any comparison table. This is a feature of the model's bookkeeping, not evidence of an error.

## 13 Case D: the model can lose its saddle path at extreme income-tax rates

**Claim to test:** "The app should be able to solve *any* combination of sliders — if it throws an error, that's a bug to file."

**Why this is (usually) wrong.** The transition path of Section 9 requires the reduced-form matrix $M$ (built from $\Phi_{kk}, \Phi_{kc}, \text{coef}_k, \text{coef}_c$) to have exactly one eigenvalue inside the unit circle and one outside — the textbook saddle-path (determinacy) condition for one predetermined and one jump variable. At sufficiently extreme income-tax rates, this condition can genuinely fail: *both* eigenvalues end up outside the unit circle, and **no bounded perfect-foresight path exists** for the linearized model. This is not a numerical bug — it is the honest answer that this particular calibration has no well-defined local equilibrium dynamics.

**Mechanism.** Under income-tax financing, a higher $G/Y$ raises $\tau$, which lowers the steady-state investment share $s_I$ (more of $Y$ is taxed away, financing government spending instead of private investment). The capital-accumulation coefficient $\Phi_{kk} = (1-\delta) + \delta\, y_k/s_I$ *rises* as $s_I$ falls — a smaller investment share means the same percentage swing in output translates into a larger swing in the capital/output ratio, amplifying the feedback loop between capital and consumption until the "stable" root of $M$ is pushed up through 1.

**Numerical example.** $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $\theta_G = 0$, all transfers ($I^G/G = 0\%$), income-tax financing. The stable eigenvalue of $M$ as $G/Y$ rises:

| $G/Y$ | $\tau$ | $N$ | $s_I$ | Stable eigenvalue of $M$ |
|---|---|---|---|---|
| 30% | 0.30 | 17.5% | 0.178 | 0.924 |
| 35% | 0.35 | 16.2% | 0.165 | 0.975 |
| 36% | 0.36 | 16.0% | 0.163 | 0.989 |
| 36.5% | 0.365 | 15.9% | 0.162 | 0.997 |
| 37% | 0.37 | 15.7% | 0.160 | 1.006 **(fails)** |

Between $G/Y = 36\%$ and 37%, the stable root crosses 1 and the saddle-path condition fails — with *zero* composition (100% transfers), this is the app's own breaking point. Raising the public-investment share pushes the threshold higher, since diverting spending into $I^G$ (rather than transfers) partially offsets the resource-cost squeeze on $s_I$'s dynamics: the same failure occurs at $G/Y \approx 39\%$ at $I^G/G = 25\%$, $\approx 43\%$ at 50%, $\approx 48\%$ at 75%, and $\approx 58\%$ at 100%.

**What the app does about it.** Rather than crashing, the app falls back to the two well-defined steady states (Section 7 never requires saddle-path stability — only the *transition* between them does) and shows a scoped warning. The long-run comparisons and long-run multiplier ($\Delta Y/\Delta G$ from the two steady states directly) remain exact and valid; only the transition-path chart, the

impact multiplier, and the duration-sensitivity replication are skipped, since they need a well-defined path from one steady state to the other. There is no single "safe" $G/Y$ cutoff to quote in class — the threshold depends on financing *and* composition jointly, as the table above shows.

## 14 Case E: "temporary" does not mean "back to the benchmark"

**Claim to test:** "If I set Duration to Temporary and shrink $G/Y$ back to 20% after a few years, S1's chart should return to exactly the benchmark, since that's where $G/Y$ started."

**Why this is wrong.** S1 compares the sidebar's current settings against the *fixed* benchmark — but financing, composition, and $\theta_G$ are always treated as a *permanent* regime switch (Section 9; there is no mechanism for a tax *rule* to revert). So when $G/Y$ reverts to 20% under a Temporary shock, the economy does *not* return to the benchmark — it settles at a *third* steady state: the *current* regime (e.g., income-tax financing) evaluated at the *benchmark's* 20% spending share. Call this $\text{ss}_{\text{reverted}}$.

**Numerical example.** $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $\theta_G = 0$, all transfers, income-tax financing, sidebar $G/Y$ moved to 30%.

| Steady state | $G/Y$ | Financing | $Y$ |
|---|---|---|---|
| Benchmark ($\text{ss}_{\text{from}}$) | 20% | Lump-sum | 0.3934 |
| Current ($\text{ss}_{\text{to}}$) | 30% | Income tax | 0.2081 |
| Reverted ($\text{ss}_{\text{reverted}}$) | 20% | Income tax | 0.2639 |

Even after $G/Y$ fully reverts to the benchmark's own 20% share, output settles 33% *below* the benchmark (0.2639 vs. 0.3934) — not because of spending, but because income-tax financing is still in effect. Reverting the *level* of spending did nothing to undo the *financing rule*.

**A numerical-methods postscript.** Naively modeling "temporary" as $G$ reverting to a nonzero constant *forever* (rather than exactly $\text{ss}_{\text{reverted}}$'s own level, i.e. $\hat{g}_t = 0$ relative to that reference) breaks the saddle-path solver's implicit terminal condition and produces an explosive, diverging path instead of convergence to $\text{ss}_{\text{reverted}}$. The fix is to linearize around $\text{ss}_{\text{reverted}}$ itself for the post-reversion phase — exactly mirroring how a temporary $\Delta G/Y$ shock in S3 already linearizes around the *old* steady state rather than the new one (Section 9). The lesson generalizes: in a perfect-foresight model, a "temporary" shock is only well-posed once you have correctly identified *where it is temporary from* — which reference point the economy actually returns to.

## 15 Case F: the real interest rate changes both the economy's scale and the fiscal multiplier

**Claim to test:** "The real interest rate $r$ (fixed at 6.5% in the app) is just the household's discount rate — it should shift the *level* of the economy up or down, but it shouldn't have anything to do with *fiscal policy*, since $r$ and $G/Y$ are two separate parameters."

**Why this is wrong.** $r$ enters the capital-Euler condition directly, $\kappa = \left(\text{tax_factor} \cdot \theta_K A/(r+\delta)\right)^{1/(1-\theta_K)}$ (Section 7): a higher $r$ raises the return households demand on capital, so the steady-

state capital/labor ratio $\kappa$ — and with it $Y$, $K$, $C$, and $w$ — fall monotonically as $r$ rises, *independent of fiscal policy entirely.* That part matches the claim. But $\kappa$ also determines the investment share $s_I = \delta\kappa/(Y/N)$, which is one of the "great ratios" feeding directly into the labor-leisure condition and, through it, into *every* fiscal-policy multiplier in S3 and S4. $r$ and $G/Y$ are separate parameters, but they are *not* separate channels in the model's mechanics.

**Numerical example — levels.** $\theta_N = 0.58$, $\delta = 10\%$, $G/Y = 20\%$, all transfers, lump-sum financing, $\theta_G = 0$, $\theta_L$ calibrated once at $r = 6.5\%$ and held fixed (Case B's convention, so $N$ is free to move honestly):

| $r$ | $\kappa$ | $Y$ | $K$ | $N$ | $s_I$ |
|---|---|---|---|---|---|
| 1.0% | 10.074 | 0.6112 | 2.3335 | 23.2% | 0.382 |
| 6.5% | 5.007 | 0.3934 | 1.0014 | 20.0% | 0.255 |
| 12.0% | 3.049 | 0.2990 | 0.5709 | 18.7% | 0.191 |

**Numerical example — the long-run multiplier** $\Delta Y/\Delta G$ (a marginal, permanent $\Delta G/Y = 1$pp increase, $\theta_L$ recalibrated at each $r$ so $N = 20\%$ at the pre-shock baseline):

| $r$ | Lump-sum, all investment | Income tax, all transfers |
|---|---|---|
| 1.0% | +1.38 | −4.57 |
| 6.5% | +1.13 | −3.92 |
| 12.0% | +1.04 | −3.68 |

Both multipliers shrink in *magnitude* as $r$ rises — the lump-sum/investment multiplier drifts down toward +1 (the point where the extra output exactly equals the extra spending, no supply-side amplification left), and the income-tax multiplier's negative effect shrinks too. The mechanism is the falling investment share $s_I$: at a high $r$, the steady state already sits in a capital-scarce, low-$\kappa$ region where investment is a smaller share of output to begin with, so the same percentage shift in labor supply or capital demand has less capital to act on and leverages a smaller output response.

**Classroom connection.** This is the model's version of a very current policy debate: fiscal multipliers are widely believed to be *larger* in low-interest-rate environments (part of the post-2008/post-2020 case for stimulus when policy rates were near zero) and smaller when rates are high. The table above shows this is not just a New Keynesian, zero-lower-bound phenomenon — it shows up even in this flexible-price, perfect-foresight real business cycle model, purely through the investment-share channel above.

## 16 Case G: longer-lasting temporary shocks have *bigger* impact multipliers

**Claim to test:** "A temporary spending increase should have a *smaller* effect on impact than a permanent one — if households know the spending burden will end soon, standard consumption-smoothing (Barro–Hall) logic says they should barely react today, and react more and more as the shock is made longer-lasting and eventually permanent, in a smoothly diminishing way that a short 1-year shock should have almost *no* impact-year effect at all, essentially zero."

**Why the direction is right but the intuition is backwards.** It is true that a longer-duration shock produces a bigger impact-year response — but the reason is the *opposite* of simple

consumption-smoothing. Consumption smoothing is about *demand*: with a short-lived burden, households can borrow/save their way through it, so consumption barely moves. But the app's multiplier is driven by the *labor-supply* response to a negative wealth effect (Case A), and the household's Euler equation (Section 7) makes today's choices forward-looking over the *entire announced path* of $g_t$: a government-spending increase that lasts 30 years is a much *larger total burden*, in present-value terms, than the same-sized increase lasting 1 year. A bigger total burden means a bigger immediate negative wealth effect, so households work *more*, right away — even though the shock hasn't actually gotten bigger *per year*, only *longer*.

**Numerical example.** $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $\theta_G = 0$, lump-sum financing, all public investment ($I^G/G = 100\%$, so the resource-cost channel of Case A is active), a marginal, temporary $\Delta G/Y$ increase of varying duration:

| Duration | Impact (year-0) multiplier $\Delta Y_0/\Delta G$ |
|---|---|
| 1 year | 0.18 |
| 2 years | 0.32 |
| 4 years | 0.52 |
| 8 years | 0.73 |
| 15 years | 0.84 |
| 30 years | 0.87 |
| Permanent ($\infty$) | 0.89 |

The long-run multiplier is 1.13 throughout this table — it is a comparative-statics number defined purely by the *two steady states* (Section 7), and does not depend on how the economy gets from one to the other, so it is the same whether or not the shock is ever actually made permanent. The impact multiplier, by contrast, rises monotonically with duration and **asymptotes to, but never exceeds, the permanent-shock impact multiplier** (0.89): a finite-duration shock is always, in present-value terms, a smaller total burden than the permanent one.

**Where to see this live.** Set item D.3 to Temporary, vary the number of years, and read the impact output multiplier in S4 (Section 9); setting D.3 to Permanent gives the permanent-shock reference value. Baxter & King make exactly this point in their Section IV, calling it the opposite of the "Barro–Hall" intuition that temporary fiscal shocks should matter less than permanent ones. It is the model's own version of "forward guidance" in modern monetary policy language: how long a policy is *expected to last* changes its effect today, independent of its per-period size.

## 17 Summary table

| Question | Answer |
|---|---|
| Does raising $I^G/G$ at $\theta_G = 0$ do anything? | **Yes** — it acts like ordinary unproductive spending (Case A). |
| Is there *any* setting where changing $G/Y$ does nothing at all? | **Yes, exactly one:** lump-sum financing *and* 100% transfers (Case A, notebox). |
| Does switching financing alone matter if all revenue is transferred back? | **Yes** — the $(1-\tau)$ wedge is on the margin, not on net income (Case B). |
| Why do $G$, $I^G$, $G^T$, $K^G$ often show identical % change? | Because $I^G = f_{I^G}G$, $G^T = f_{G^T}G$, and $K^G = I^G/\delta$ are exact identities whenever composition shares and $\delta$ are unchanged (Case C). |
| Why does the app sometimes refuse to compute a transition path? | Extreme income-tax rates (with a low investment share) can push the model's saddle path to lose determinacy — a genuine breakdown of the linearized dynamics, not a bug (Case D). |
| Does a "temporary" G/Y increase return the economy to the benchmark? | **No** — only spending reverts; financing/composition/$\theta_G$ don't, so the economy settles at a third steady state, not the original benchmark (Case E). |
| Does the real interest rate $r$ affect fiscal multipliers, or only levels? | **Both** — $r$ sets $\kappa$ and hence the investment share $s_I$, which feeds directly into every multiplier; higher $r$ shrinks multiplier magnitude (Case F). |
| Should a longer-lasting temporary shock have a *smaller* impact effect (consumption-smoothing intuition)? | **No** — the *opposite*: longer duration means a bigger present-value burden, so the labor-supply response (and hence the impact multiplier) rises monotonically with duration, asymptoting to the permanent-shock value (Case G). |

# Part III
# Appendix: full derivations

This appendix derives, from scratch, every equation asserted without proof in Part I: the two household first-order conditions (labor-leisure and Euler) for both financing rules, and the closed-form steady-state equations built on top of them. Nothing here is new economics — it is the algebra that Part I skipped over.

## A The firm's problem (why $w_t = \mathrm{MPL}_t$ and $r_t = \mathrm{MPK}_t$)

Before touching the household, it is worth pinning down where the prices $w_t$ and $r_t$ in the household's budget constraint actually come from. A representative, perfectly competitive firm rents private capital and labor from households each period and uses the publicly provided capital stock $K_t^G$ for free (it is non-rival and supplied by the government, not purchased in a market). The firm solves, period by period,

$$\max_{K_t,\, N_t} \quad A\, K_t^{\theta_K} (K_t^G)^{\theta_G} N_t^{\theta_N} \;-\; w_t N_t \;-\; r_t K_t. \tag{24}$$

Since $K_t^G$ is taken as given (external to the firm's own choice), the first-order conditions are the ordinary marginal-product conditions:

$$\frac{\partial Y_t}{\partial N_t} = w_t \implies w_t = \theta_N \frac{Y_t}{N_t} \equiv \text{MPL}_t, \tag{25}$$

$$\frac{\partial Y_t}{\partial K_t} = r_t \implies r_t = \theta_K \frac{Y_t}{K_t} \equiv \text{MPK}_t. \tag{26}$$

Zero profit follows automatically from constant returns to scale in $(K_t, N_t)$ (Euler's homogeneous-function theorem): at these prices the firm earns exactly $\theta_K Y_t + \theta_N Y_t = Y_t$, paying out all of output to the two rented factors. This is why $r_t$ (the household's capital income rate) and $\text{MPK}_t$ can be used interchangeably below — they are the *same number* in every period, by the firm's optimality condition, not by assumption.

# B Setting up the household's problem

The household chooses sequences $\{C_t, N_t, K_{t+1}\}_{t=0}^{\infty}$ to maximize discounted lifetime utility,

$$\max \sum_{t=0}^{\infty} \beta^t \Big[ \ln C_t + \theta_L \ln(1 - N_t) \Big], \qquad \beta = \frac{1}{1+r}, \tag{27}$$

subject to one budget constraint per period and the capital accumulation identity $K_{t+1} = (1 - \delta)K_t + I_t$, i.e. $I_t = K_{t+1} - (1-\delta)K_t$. Substituting this into the two budget constraints of Section 5 gives, respectively,

$$\text{Income tax:} \qquad C_t + K_{t+1} - (1-\delta)K_t = (1-\tau_t)\big(w_t N_t + r_t K_t\big) + G_t^T, \tag{28}$$

$$\text{Lump-sum:} \qquad C_t + K_{t+1} - (1-\delta)K_t = w_t N_t + r_t K_t - \tau_t Y_t + G_t^T. \tag{29}$$

There is no expectation operator anywhere in this problem: the model has no stochastic element, so the household is choosing a sequence under **perfect foresight** — it knows $w_t, r_t, \tau_t, G_t^T$ (and hence $Y_t$) at every future date along the equilibrium path, given the policy being analyzed. Two modeling conventions are worth flagging explicitly:

(i) Under *income tax* financing, the wedge $(1 - \tau_t)$ multiplies the household's *own* factor income $w_t N_t + r_t K_t$ — it is a genuine proportional tax on the household's choices, so it will show up when we differentiate with respect to $N_t$ and $K_{t+1}$ below.

(ii) Under *lump-sum* financing, the household's tax bill is written as $\tau_t Y_t$, using *aggregate* output $Y_t$, not the household's own income. The (representative) household takes $\tau_t Y_t$ as a parameter it cannot affect with its own $N_t$ or $K_t$ choice — exactly what makes a lump-sum tax lump-sum. This is why $\tau_t$ never appears in the lump-sum first-order conditions below, even though $\tau_t Y_t > 0$ in general.

# C Deriving the labor-leisure and Euler conditions

Attach a Lagrange multiplier $\lambda_t \geq 0$ (period-$t$'s marginal utility of an extra unit of income) to the period-$t$ budget constraint and form the Lagrangian. Under **income tax** financing,

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \ln C_t + \theta_L \ln(1 - N_t) + \lambda_t \Big[ (1-\tau_t)(w_t N_t + r_t K_t) + G_t^T - C_t - K_{t+1} + (1-\delta)K_t \Big] \Big\}. \tag{30}$$

**Step 1: the consumption FOC.** Differentiating with respect to $C_t$ (which appears only in period $t$'s bracket) gives

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t \left[ \frac{1}{C_t} - \lambda_t \right] = 0 \implies \lambda_t = \frac{1}{C_t}. \tag{31}$$

This is the standard result that the multiplier on the budget constraint equals the marginal utility of consumption — an extra unit of income is worth exactly what an extra unit of consumption is worth, since the household can always convert one into the other one-for-one within a period.

**Step 2: the labor-leisure FOC.** Differentiating with respect to $N_t$ (also period-$t$-only) gives

$$\frac{\partial \mathcal{L}}{\partial N_t} = \beta^t \left[ -\frac{\theta_L}{1 - N_t} + \lambda_t (1 - \tau_t) w_t \right] = 0. \tag{32}$$

(The $-\theta_L/(1 - N_t)$ term has a sign flip built in: $\partial \ln(1 - N_t)/\partial N_t = -1/(1 - N_t)$.) Solving and substituting $\lambda_t = 1/C_t$ from Step 1,

$$\frac{\theta_L}{1 - N_t} = \frac{(1 - \tau_t) w_t}{C_t} \implies \boxed{\theta_L \, \frac{C_t}{1 - N_t} = (1 - \tau_t) \, w_t.} \tag{33}$$

This is exactly the labor-leisure condition of Section 6. Economically: the left side is the marginal rate of substitution between consumption and leisure (how many units of consumption the household would give up for one more unit of leisure); the right side is the after-tax wage, the market price of leisure in consumption units. Optimality equates the two.

**Step 3: the Euler (capital) FOC.** $K_{t+1}$ is the one variable that appears in *two* budget constraints: negatively in period $t$'s (it is being accumulated) and positively in period $t + 1$'s (it earns capital income and is then reduced by depreciation and reinvestment). Collecting both appearances,

$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = \beta^t (-\lambda_t) + \beta^{t+1} \lambda_{t+1} \Big[ (1 - \tau_{t+1}) r_{t+1} + (1 - \delta) \Big] = 0. \tag{34}$$

Dividing through by $\beta^t$ and rearranging,

$$\lambda_t = \beta \, \lambda_{t+1} \Big[ 1 + (1 - \tau_{t+1}) r_{t+1} - \delta \Big]. \tag{35}$$

Substituting $\lambda_t = 1/C_t$, $\lambda_{t+1} = 1/C_{t+1}$, and $r_{t+1} = \mathrm{MPK}_{t+1}$ (Appendix A),

$$\boxed{\frac{1}{C_t} = \beta \, \frac{1 + (1 - \tau_{t+1}) \, \mathrm{MPK}_{t+1} - \delta}{C_{t+1}}.} \tag{36}$$

This is exactly the income-tax Euler equation of Section 6. The bracketed term $1 + (1 - \tau_{t+1})\mathrm{MPK}_{t+1} - \delta$ is the after-tax gross return on one unit of capital held from $t$ to $t + 1$: it pays $(1 - \tau_{t+1})\mathrm{MPK}_{t+1}$ in after-tax rental income and loses $\delta$ to depreciation, on top of the principal. The household is indifferent at the margin between consuming a unit today (worth $1/C_t$ in utility, by Step 1) and investing it to consume the gross return tomorrow (worth $\beta \times \text{return} \times 1/C_{t+1}$ in discounted utility).

**The lump-sum case.** Repeat Steps 2–3 with the lump-sum Lagrangian,

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \ln C_t + \theta_L \ln(1 - N_t) + \lambda_t \Big[ w_t N_t + r_t K_t - \tau_t Y_t + G_t^T - C_t - K_{t+1} + (1-\delta) K_t \Big] \Big\}. \quad (37)$$

Because $\tau_t Y_t$ does not depend on $N_t$ or $K_{t+1}$ (point (ii) of Appendix B), it simply drops out when differentiating:

$$\frac{\partial \mathcal{L}}{\partial N_t} = \beta^t \left[ -\frac{\theta_L}{1 - N_t} + \lambda_t w_t \right] = 0 \implies \boxed{\theta_L \frac{C_t}{1 - N_t} = w_t,} \quad (38)$$

$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = \beta^t(-\lambda_t) + \beta^{t+1} \lambda_{t+1} \big[ r_{t+1} + (1-\delta) \big] = 0 \implies \boxed{\frac{1}{C_t} = \beta \frac{1 + \mathrm{MPK}_{t+1} - \delta}{C_{t+1}}.} \quad (39)$$

No $(1-\tau)$ wedge appears in either condition: the household's marginal trade-offs between consumption/leisure and consumption today/tomorrow are completely undistorted, even though the household *does* pay $\tau_t Y_t$ in taxes every period. This is precisely the sense in which a lump-sum tax is non-distorting — it changes the household's wealth, but not the *price* of working an extra hour or saving an extra dollar.

# D Deriving the closed-form steady state

A steady state is a situation where every "great ratio" and every level variable is constant over time: $C_t = C$, $N_t = N$, $K_t = K$, $Y_t = Y$, and so on, for all $t$. This section derives every equation of Section 7 in order, starting from the FOCs of Appendix C.

## D.1 Step 1: the capital-Euler condition pins down $\kappa = K/N$

In steady state $C_t = C_{t+1} = C$ and $\mathrm{MPK}_t = \mathrm{MPK}_{t+1} = \mathrm{MPK}$, so the Euler equation collapses to an equation with no time subscripts at all. Income tax case:

$$1 = \beta \big[ 1 + (1-\tau)\mathrm{MPK} - \delta \big] \implies (1-\tau)\mathrm{MPK} = \frac{1}{\beta} - 1 + \delta. \quad (40)$$

Recall $\beta = 1/(1+r)$ was *chosen* so that $1/\beta = 1 + r$ (the household's steady-state discount rate is calibrated to match a target real interest rate $r$). Substituting,

$$\boxed{(1-\tau)\,\mathrm{MPK} = r + \delta.} \quad (41)$$

The lump-sum case is identical without the $(1-\tau)$ factor: $\boxed{\mathrm{MPK} = r + \delta.}$ These are the two boxed conditions quoted without proof at the start of Section 7.

Now expand $\mathrm{MPK} = \theta_K Y/K$ using the technology equation, and substitute $K = \kappa N$ (definition of $\kappa$):

$$\mathrm{MPK} = \theta_K \frac{A K^{\theta_K} (K^G)^{\theta_G} N^{\theta_N}}{K} = \theta_K A\, K^{\theta_K - 1} (K^G)^{\theta_G} N^{\theta_N} = \theta_K A\, \kappa^{\theta_K - 1} N^{\theta_K - 1} (K^G)^{\theta_G} N^{\theta_N}. \quad (42)$$

Since $\theta_K + \theta_N = 1$ (constant returns to scale in $(K, N)$ alone, Section 4), the two powers of $N$ combine to $N^{\theta_K - 1 + \theta_N} = N^0 = 1$, leaving

$$\mathrm{MPK} = \theta_K A\, \kappa^{\theta_K - 1}\, (K^G)^{\theta_G}. \quad (43)$$

Notice this uses the *raw* level $K^G$, not $K^G/N$: no $N$ survives the cancellation above. To match the per-worker form used in Section 7 and in the app, multiply and divide by $N^{\theta_G}$:

$$(K^G)^{\theta_G} = \left(\frac{K^G}{N}\right)^{\theta_G} N^{\theta_G} \implies \text{MPK} = \theta_K A \left(\frac{K^G}{N}\right)^{\theta_G} N^{\theta_G}\, \kappa^{\theta_K - 1}. \tag{44}$$

Substituting into the boxed conditions above reproduces exactly

$$\text{Income tax:} \quad (1-\tau)\,\theta_K\, A\,(K^G/N)^{\theta_G}\, N^{\theta_G}\, \kappa^{\theta_K - 1} = r + \delta, \tag{45}$$

$$\text{Lump-sum:} \quad \theta_K\, A\,(K^G/N)^{\theta_G}\, N^{\theta_G}\, \kappa^{\theta_K - 1} = r + \delta. \tag{46}$$

**Why bother with the $(K^G/N)^{\theta_G} N^{\theta_G}$ split instead of just writing $(K^G)^{\theta_G}$?** Because in the fixed-point iteration the app actually solves numerically, $K^G$ is not known in advance when $\theta_G > 0$ — it depends on output through $K^G = I^G/\delta$, which depends on $\kappa$, which is what we are solving for. Working in per-worker terms $(K^G/N)$ lets the iteration guess and update a *scale-free ratio* first, and then multiply back in the explicit $N^{\theta_G}$ correction, rather than guessing the level $K^G$ directly. When $\theta_G = 0$, $N^{\theta_G} = 1$ and this distinction is invisible; it only matters once public capital is productive.

When $\theta_G = 0$, equation (43) has no $K^G$ dependence at all, and can be solved for $\kappa$ directly:

$$\theta_K A\, \kappa^{\theta_K - 1} = \frac{r+\delta}{\text{tax_factor}} \implies \kappa = \left(\frac{\text{tax_factor} \cdot \theta_K A}{r+\delta}\right)^{\frac{1}{1-\theta_K}}, \tag{47}$$

where tax_factor $= (1-\tau)$ under income tax and 1 under lump-sum. When $\theta_G > 0$, the same formula holds but with $A$ replaced by $A\,(K^G/N)^{\theta_G} N^{\theta_G}$, which itself depends on $\kappa$ through $K^G = I^G/\delta$ and $Y = \alpha N$ — hence the nested fixed-point iteration described in Section 7.

### D.2 Step 2: the investment and consumption shares

With $\kappa$ (and hence $Y/N \equiv \alpha$) in hand, define the steady-state investment share directly from the capital accumulation identity $K_{t+1} = (1-\delta)K_t + I_t$: in steady state $K_{t+1} = K_t = K$, so $I = \delta K$. Dividing by $Y$,

$$s_I \equiv \frac{I}{Y} = \frac{\delta K}{Y} = \frac{\delta \kappa N}{\alpha N} = \frac{\delta \kappa}{\alpha}. \tag{48}$$

The resource constraint $Y = C + I + I^G$ (Section 5), divided by $Y$, gives $1 = s_C + s_I + s_{I^G}$, i.e.

$$\boxed{s_C = 1 - s_I - s_{I^G}.} \tag{49}$$

Transfers $G^T$ never appear in this identity, because they were never in the resource constraint to begin with (Section 5) — this is the formal statement of "transfers are not resource-using."

### D.3 Step 3: the labor-leisure condition pins down $N$

Start from the steady-state labor-leisure condition (income tax case),

$$\theta_L \frac{C}{1-N} = (1-\tau)\, w, \qquad w = \theta_N \frac{Y}{N} \ \text{(Appendix A)}. \tag{50}$$

Multiply both sides by $(1 - N)$ and divide both sides by $Y$, using $C/Y = s_C$:

$$\theta_L \, s_C = (1 - \tau) \, \theta_N \, \frac{1 - N}{N}. \tag{51}$$

Multiply both sides by $N$:

$$\theta_L \, s_C \, N = (1 - \tau) \, \theta_N \, (1 - N) = (1 - \tau)\theta_N - (1 - \tau)\theta_N \, N. \tag{52}$$

Collect every term containing $N$ on the left:

$$N\big[\theta_L \, s_C + (1 - \tau)\theta_N\big] = (1 - \tau)\theta_N. \tag{53}$$

Divide through:

$$\boxed{N = \frac{(1 - \tau) \, \theta_N}{\theta_L \, s_C + (1 - \tau) \, \theta_N}.} \tag{54}$$

The lump-sum case follows identically with tax_factor $= 1$ in place of $(1 - \tau)$ throughout (there is no wedge in the lump-sum labor-leisure condition to begin with):

$$\boxed{N = \frac{\theta_N}{\theta_L \, s_C + \theta_N}.} \tag{55}$$

These reproduce exactly the two boxed $N$-equations of Section 7. Notice the entire derivation only used *ratios* ($s_C = C/Y$, $\theta_L, \theta_N, \tau$) — this is why $N$ can be solved before $Y$, $C$, or $K$ are known in levels: once $\kappa$ fixes $\alpha = Y/N$ and $s_C$ (Steps 1–2), $N$ falls out of a single algebraic equation, and every level variable is then a trivial multiple of $N$:

$$Y = \alpha N, \qquad K = \kappa N, \qquad C = s_C \, Y, \qquad I = s_I \, Y, \qquad I^G = s_{I^G} \, Y, \qquad G^T = s_{G^T} \, Y. \tag{56}$$

**Where does $\theta_L$ come from?** $\theta_L$ is not observed directly; it is *calibrated* by inverting the boxed $N$-equation at a chosen baseline policy and a target hours-worked share $N_{\text{target}}$ (Baxter & King's Table 1 strategy). Solving the income-tax $N$-equation for $\theta_L$ instead of $N$:

$$\theta_L = \frac{(1 - \tau) \, \theta_N \, (1 - N_{\text{target}})}{s_C \, N_{\text{target}}}, \tag{57}$$

and analogously without the $(1-\tau)$ factor under lump-sum. This is exactly `calibrate_theta_L` in the code: it is the *same* equation as Step 3 above, just solved for the other unknown.